

Description

This week, I learned how supervised machine learning models are fitted by defining an error metric and minimising the corresponding cost function. The focus was linear regression, where I compared ordinary least squares (OLS) with gradient descent. The mentor meeting also helped me reflect on my development as a future data scientist.

Feelings

At first, I felt slightly overwhelmed by the mathematical concepts behind optimisation, especially gradient descent and OLS. However, I became more confident after understanding that both methods aim to minimise prediction error.

Evaluation

My technical strength is that I can understand the purpose of error metrics and cost functions and apply these concepts using programming tools. However, I need to improve my understanding of the mathematical reasoning behind algorithms rather than relying mainly on built-in functions. This limitation may make it more difficult for me to interpret model results or identify problems when I do not fully understand how the algorithm works.

Analysis

During the mentor meeting, I realised that explaining my reasoning and technical decisions clearly is as important as producing correct code. This is because data scientists need to justify their approaches to others, while clear communication also allows others to understand my reasoning and provide useful feedback.

Conclusion

Overall, this week showed me that becoming a data scientist requires both technical knowledge and continuous self-improvement. I have developed a better understanding of how optimization is used to fit machine learning models, but I still need to strengthen my mathematical understanding and communication skills. Looking back, I could have spent more time reviewing the mathematical concepts before applying them in the notebook.

Action plan

In future, I will implement simple algorithms from scratch before using scikit-learn and compare the results. I will spend 30 minutes each week reviewing the mathematical steps behind machine learning algorithms. Before mentor meetings, I will prepare at least two questions and explain my approach to practise communicating technical ideas clearly.